ASSIGNMENT NO 3

Hostel Accommodation Portal

Team members: 25L-5223 and 25L-5215

1. Executive Summary

The Hostel Accommodation Portal is a console program written in Python. It helps a hostel warden keep track of rooms and students instead of using a paper register.

Purpose: the warden can log in, add rooms and students, search for a student, give a room to a student, vacate a room and see a report of how many beds are occupied.

Scope: the project is backend only and uses a text menu. All data is stored in two CSV files, so records are still available when the program is started again.

2. System Architecture

2.1 Functions

Function	What it does
user_login(username, password)	Checks the username and password. Returns True or False.
load_file(filename)	Reads a CSV file and returns the rows as a list.
save_file(filename, data)	Writes the list of rows back to the CSV file.
add_room(rooms, room_no, capacity, rent)	Checks the input and adds a new room. Returns a message.
add_student(students, roll_no, name)	Adds a new student. Does not allow a repeated roll number.
search_record(records, keyword)	Searches the rows using a for loop and returns the matches.
show_records(records, heading)	Prints rows on the screen.
allocate_room(...) / vacate_room(...)	Gives a student a room (if it is not full) or removes the student from it.
generate_report(students, rooms)	Prints total rooms, students, occupied and free beds, occupancy percentage and students without a room.
main()	Login (3 tries) and the while True menu loop.

2.2 Loops

- A while True loop shows the menu again and again until the user enters 0.
- A while loop gives the user 3 tries to log in.

- for loops are used to search records, find a room or student, show records and count beds in the report.

2.3 File structure

File	Columns
rooms.csv	room_no, capacity, occupied, rent
students.csv	roll_no, name, room_no (empty if the student has no room)

The files are read when the program starts and saved again after every change.

2.4 Error handling

Problem	Exception	What the program does
CSV file not found	FileNotFoundError	Shows a message and starts with an empty list. The file is created on the first save.
File cannot be opened or saved	PermissionError	Shows a message and keeps running.
Letters typed in capacity or rent	ValueError	Shows a message and does not add the room.
Wrong menu choice	if / else	Asks the user to enter a valid number.

In save_file the finally block always closes the file, even when an error happens.

3. Testing

Test	Result
Wrong password, then correct password	First try rejected, second accepted.
Letters typed as capacity	Message shown, room not added.
Same roll number added twice	Rejected with a message.
Allocating a student to a full room	Message: Room R201 is full.
Missing CSV file at start	Message shown, program continues.
Invalid menu choice	Message shown, menu appears again.

4. Work Division Matrix

Member	Backend part	Documentation part
25L-5223	user_login, load_file, save_file, add_room, add_student	Executive summary, file structure, error handling

Member	Backend part	Documentation part
25L-5215	search_record, show_records, allocate_room, vacate_room, generate_report	Function table, testing, README
Both	main() menu loop, Jupyter notebook, GitHub upload	Screenshots and PDF export

5. Conclusion

The portal uses functions, loops, CSV file handling and exception handling as required by the assignment. In the future a fee system and a student login can be added.